Class Imbalance in Histopathology Patch Classification

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September 25, 2026

1 Motivation

Histopathology datasets are long-tailed almost by construction. Common cancers and benign tissue dominate archives, while rare subtypes and atypical lesions, often diagnostically decisive, contribute few patients and patches. In convolutional neural networks, imbalance degrades classification, and the network absorbs the training class prior (Buda et al., 2018). In medical imaging, metrics dominated by the majority class can hide poor minority-class performance in both discrimination and calibration (Mosquera et al., 2024). Researchers continue to propose histopathology-specific models (Cong et al., 2022). Pathology foundation models such as UNI and Virchow2 are commonly evaluated with linear classifiers trained on their frozen patch features (Chen et al., 2024; Zimmermann et al., 2024). Yet how much class imbalance costs such classifiers, and why, remains insufficiently measured in isolation.

Three questions guide this work. How much do discrimination and calibration degrade as the training class distribution becomes more skewed (Section 3)? What causes this damage, and why are some classes and tasks more vulnerable than others (Section 4)? How much of the damage can mitigation methods recover (Section 5)?

2 Setup

Multinomial logistic regression on Virchow2 patch features (Zimmermann et al., 2024) addresses two tasks: TCGA-UT, 30 cancer types (Komura et al., 2022), and BRACS, 7 breast-lesion subtypes from normal tissue to invasive carcinoma (Brancati et al., 2022). Allocation conditions use the same patients per class (20 on TCGA-UT, 10 on BRACS) and total patch budget; only class shares change. An exponential profile sets the largest-to-smallest class ratio ρ from 1 to 100 (Figure 1). Class-to-rank assignment is randomized over 30 draws (3 splits $\times$ 10 draws); a native condition preserves each dataset's class shares. The outcome is *damage*, $\Delta(\rho) = \mathrm{BA}(1) - \mathrm{BA}(\rho)$: the loss in test patient-macro balanced accuracy (BA) relative to the balanced condition, with paired bootstrap 95% intervals. Damage counts as material from 1 percentage point (pp).

The encoder stays frozen when measuring damage and its causes. Mitigation methods that adapt the encoder fine-tune it through low-rank adaptation (LoRA) (Hu et al., 2022). To rule out dependence on one encoder, the conditions at $\rho = 10$ and $\rho = 100$ are repeated on the same patients and patches with UNI2-h, the successor of UNI (Mahmood Lab, 2025).

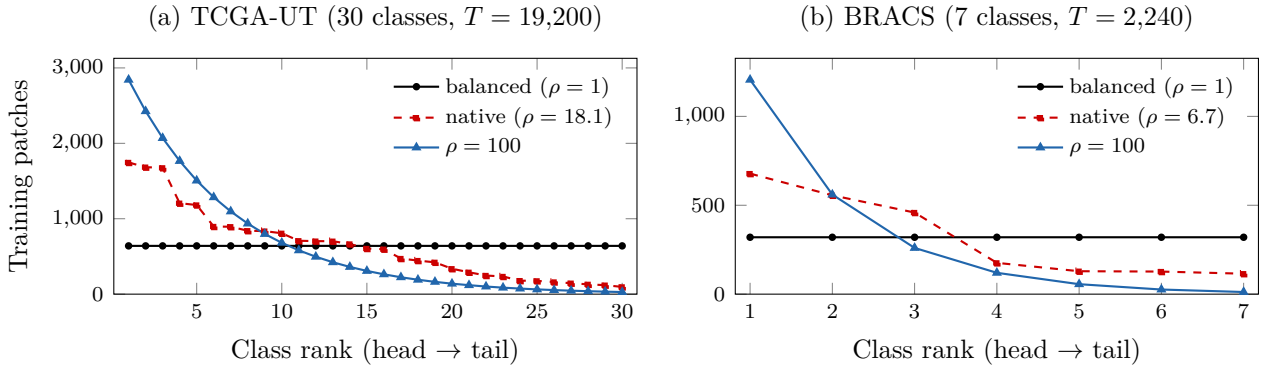


Figure 1: Training class distributions sorted from head to tail: balanced, native shares of one split, and $\rho = 100$. The tested ratios are $\rho \in \{1, 2, 5, 10, 20, 50, 100\}$; intermediate profiles lie between balanced and $\rho = 100$. Class-to-rank assignment is randomized per draw. T: total training patch budget.

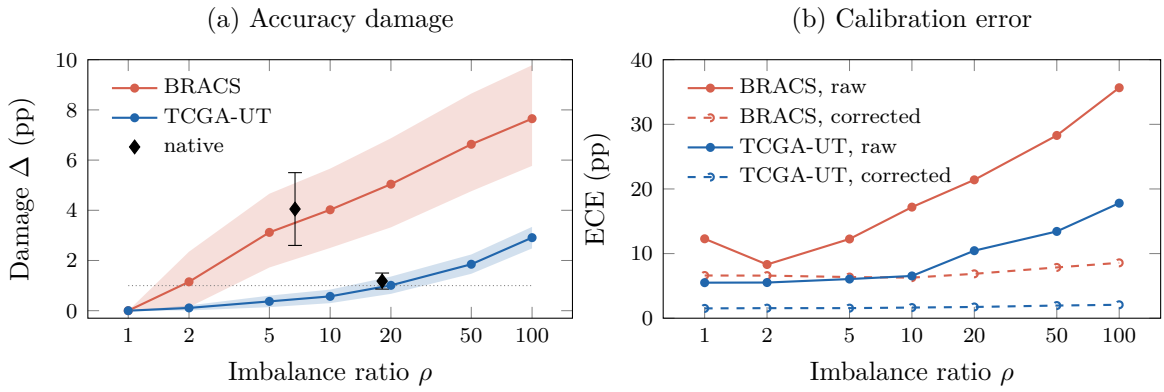


Figure 2: (a) Balanced-accuracy damage versus ρ with 95% bands; diamonds mark native class shares, dotted line the 1-pp threshold. (b) Expected calibration error (ECE) of the raw predicted probabilities (solid) and after a single global confidence correction fitted on validation data (dashed).

3 Effects of class imbalance

Balanced accuracy in the balanced condition reaches 72.1% on TCGA-UT and 39.6% on BRACS. Published evaluations show the same broad gap: region-of-interest (ROI)-level benchmarks report 59.5–67.5% for UNI, Virchow2, and UNI2-h on the 32-class TCGA-UT task (Mahmood Lab, 2025), while slide-level linear probes score 31.8–48.1% on BRACS (Vaidya et al., 2025). TCGA-UT spans cancer types across organs, while BRACS separates breast lesions with similar morphology, such as atypical ductal hyperplasia (ADH) and ductal carcinoma in situ (DCIS) (Brancati et al., 2022). These evaluations make the gap plausible, though their training budgets and evaluation units differ. This study computes patient-macro balanced accuracy from patch predictions.

Imbalance damage differs strongly between the two tasks (Figure 2a). On TCGA-UT, damage is small but accelerates as ρ increases: 0.57 pp [0.30, 0.84] at $\rho = 10$, reaching the 1-pp threshold at $\rho = 20$ (1.01 pp [0.67, 1.36]) and 2.91 pp [2.48, 3.34] at $\rho = 100$. On BRACS, damage appears at $\rho = 2$ and grows steadily by about 1.15 pp per doubling of ρ: 1.15 pp [0.14, 2.35] at $\rho = 2$, 4.02 pp [2.50, 5.66] at $\rho = 10$, and 7.65 pp [5.77, 9.78] at $\rho = 100$, a fifth of the balanced accuracy. The native allocations lie on their respective curves (1.18 pp [0.85, 1.48] on TCGA-UT and 4.05 pp [2.69, 5.57] on BRACS). Aggregate damage hides a large shift in recall from tail to head. At $\rho = 100$, the TCGA-UT tail third loses 11.8 pp while the head gains 5.4 pp; on BRACS, the two tail classes drop from 38.7% to 7.3% recall.

Calibration deteriorates far more than accuracy (Figure 2b). Expected calibration error (ECE), the average gap between the model's confidence and its accuracy, rises from 5.5 to 17.8 pp on TCGA-UT and from 12.3 to 35.7 pp on BRACS at $\rho = 100$: the model becomes overconfident, but uniformly so. Temperature scaling divides all logits by one scalar fitted on validation data (Guo et al., 2017). This global confidence correction removes 92–96% of the increase.

Replacing Virchow2 with UNI2-h changes the damage little. On TCGA-UT, UNI2-h's balanced baseline is 1.78 pp higher, and its damage at $\rho = 100$ is 0.86 pp lower (97.5% interval 0.40 to 1.33): a resolved difference, but below the 1-pp threshold. On BRACS, the difference is unresolved, and tail-class recall falls to about 9% with both encoders.

4 Causes of class imbalance damage

Two questions explain the damage. The first is which changes in the training data cause it (Section 4.1). The second is why the same imbalance ratio costs more for some rare classes than for others (Section 4.2).

4.1 Sources of damage

When a tail class receives fewer training patches, two things change. Its errors contribute less often to the training loss, so common classes have more influence on the fitted decision boundary. This is the

frequency effect: how strongly each class's available examples shape the classifier. The model also sees fewer distinct tail patches, only one or two per patient at $\rho = 100$, so it has less evidence about how that class varies. This is the *support* effect: giving those few patches more weight cannot supply the missing examples.

The conditions in Table 1 separate these effects. F keeps all patches from the balanced condition but gives head classes more weight, isolating the frequency effect. S keeps the sparse patches from the imbalanced condition but weights classes equally, isolating the support effect. The ratio condition r combines both effects. The interaction $I = \Delta_r - \Delta_F - \Delta_S$ measures any additional damage when both occur together.

Class influence	Many tail patches	Few tail patches
Equal	balanced (ref.)	S: support only
Head-favoured	F: frequency only	r: ratio (both)

Table 1: Four training conditions separate the effect of unequal class influence (frequency) from the effect of fewer tail examples (support) at each ρ.

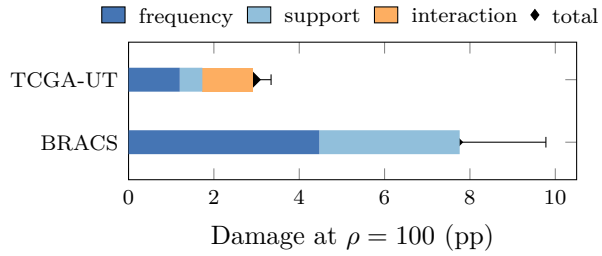


Figure 3: Damage at $\rho = 100$ by component; diamonds give the total with its 95% interval.

The damage components differ between tasks (Figure 3). On TCGA-UT, the frequency effect accounts for damage at moderate ratios (0.73 of 1.01 pp at $\rho = 20$), while sparse tail support alone costs nothing. At $\rho = 100$, the two effects compound: frequency 1.20 pp [0.87, 1.53], support 0.53 pp [0.36, 0.70], and a positive interaction of 1.18 pp [1.00, 1.37]. On BRACS, both factors are material at $\rho = 100$: frequency 4.47 pp [2.84, 6.30] and support 3.29 pp [1.99, 4.67]. Their sum matches the total damage. The interaction is near zero, −0.11 pp, but its interval [−1.74, 1.36] cannot rule out compounding of up to 1.4 pp. Sparse tail support alone costs BRACS more than total TCGA-UT imbalance damage.

4.2 Vulnerable classes

On BRACS, rotating every class through every rank shows that easy classes cost most when rare. Damage at $\rho = 100$ ranges from 5.73 pp (usual ductal hyperplasia, UDH, in the tail) to 9.99 pp (invasive carcinoma), because a class with high balanced recall has the most to lose (Figure 4a).

On TCGA-UT, a class is especially vulnerable when it receives few patches but a frequent confusion partner receives many. A score fixed before testing new class-to-rank assignments predicted their damage from this pattern. The assignment ranked most damaging costs 4.51 pp, the least damaging 1.34 pp, and random assignments 2.91 pp (Figure 4b). Reversing the most damaging assignment preserves pairwise rank gaps but lowers damage by 1.11 pp [0.47, 1.73]. Damage is therefore greatest when the class prone to confusion is the rarer one.

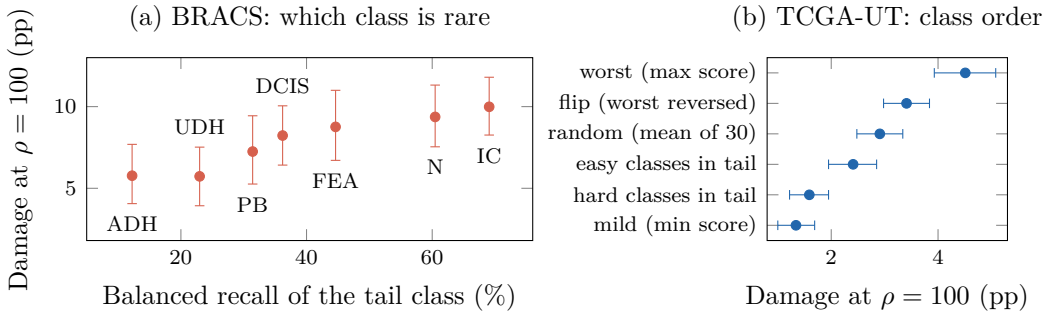


Figure 4: Damage at $\rho = 100$ with 95% intervals. (a) BRACS tail-class identity versus balanced recall (Spearman correlation 0.96, $n = 7$). ADH: atypical ductal hyperplasia; UDH: usual ductal hyperplasia; PB: pathological benign; DCIS: ductal carcinoma in situ; FEA: flat epithelial atypia; N: normal; IC: invasive carcinoma. (b) TCGA-UT class-to-rank assignments.

5 Mitigation of class imbalance damage

Section 4 traced the damage to the frequency and support effects. This section asks how much of it established long-tailed methods recover compared with standard cross-entropy (CE) training. Six methods cover the three families of Krawczyk (2016) (Table 2). Data-level methods change the training data, algorithm-level methods change the loss or the decision rule, and hybrid methods do both.

Several of these methods act on the learned representation, not only on the classifier. The encoder is therefore adapted through LoRA together with a linear head. The patients, patch budgets, and class-to-rank assignments match those in Section 3. CE and each method are trained at $\rho = 100$, $\rho = 50$, and native shares, and a CE model trained on the balanced allocation provides the undamaged reference. Each method except cRT has one control parameter that sets how strongly it acts, such as the scale of the logit adjustment or the probability that CUDA augments a patch; it is selected per draw on validation patients.

Under LoRA, CE loses 4.95 pp [3.25, 6.74] on BRACS and 1.56 pp [1.17, 1.94] on TCGA-UT at $\rho = 100$. A method's *recovery* is its gain in balanced accuracy over CE at the same allocation; its *share* is recovery divided by CE's damage. Methods whose effect on balanced data is not zero by construction are also fitted on the balanced allocation. Recovery minus this *balanced gain* is the part that corrects the imbalance. Results refer to Virchow2 and $\rho = 100$ unless stated otherwise.

Family	Method	What changes
Data	Balanced sampling (Kang et al., 2020)	Every class is drawn equally often.
	CUDA (Ahn et al., 2023)	Image patches are augmented per class, more strongly once the model has learned the class.
Algorithm	Logit adjustment (Menon et al., 2021)	The training class prior is subtracted from the class scores, in the loss or after training.
	DisAlign (Zhang et al., 2021)	A loss that weights rare classes more trains a per-class score correction.
Hybrid	cRT (Kang et al., 2020)	The head is refitted with balanced sampling.
	GCL (Li et al., 2022)	Random noise lowers the scores of rarer classes more during training; optionally, the head is then refitted with near-balanced sampling.

Table 2: The six mitigation methods change the data, the learner, or both. Logit adjustment is tested in the loss and after training, GCL with and without the head refit, giving eight variants. CUDA: curriculum of data augmentation; DisAlign: distribution alignment; cRT: classifier re-training; GCL: Gaussian clouded logit adjustment.

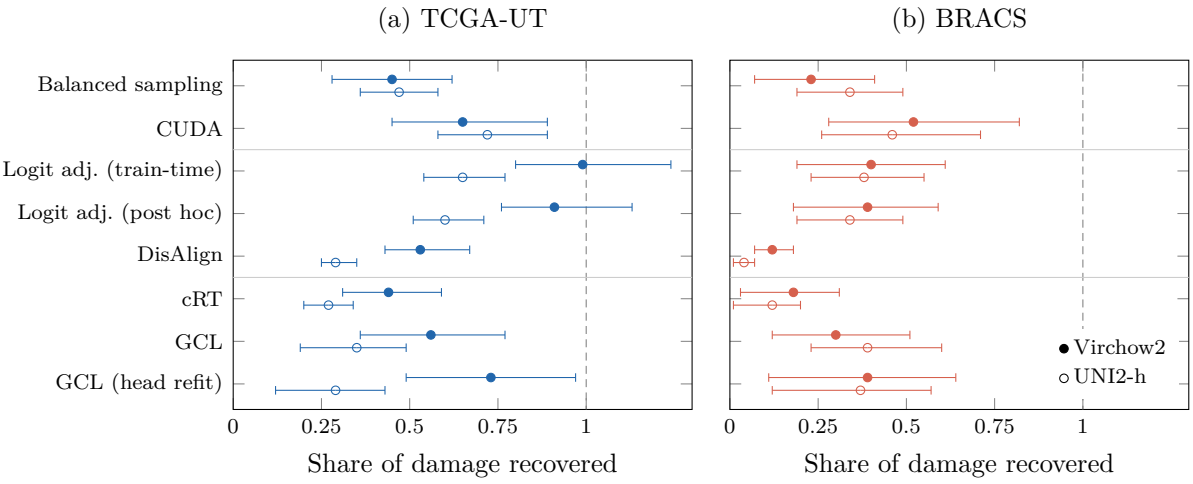


Figure 5: Only logit adjustment on TCGA-UT with Virchow2 recovers the full damage; on BRACS no method recovers much more than half of it. Each method's share of CE damage recovered at $\rho = 100$, with paired 95% intervals, for Virchow2 (filled) and UNI2-h (hollow) on the same patients and patches. The dashed line marks full recovery. Grey lines separate the data-level, algorithm-level, and hybrid families of Table 2.

5.1 What mitigation achieves

1. Cheap fixes already recover much of the damage. Logit adjustment removes the training prior from the class scores (Menon et al., 2021). Applied to the CE model after training, it needs no retraining and recovers as much as its train-time form: 1.41 against 1.54 pp on TCGA-UT and 1.94 against 1.96 pp on BRACS. On TCGA-UT this is essentially the full damage (Figure 5a). The train-time share is 0.99 [0.80, 1.24], and recall of the tail third of classes rises from 61.2% to 69.4%, against 71.6% when balanced. On BRACS, both forms recover about two-fifths (Figure 5b). Balanced sampling, which changes only how often each class is drawn, recovers at least 1 pp in three of the four dataset–encoder settings.

2. Augmentation improves the model in general. CUDA recovers 0.46 to 0.72 of the damage across the four settings, the largest share in all but TCGA-UT with Virchow2, where logit adjustment recovers more. Most of this gain does not depend on imbalance: CUDA also improves the balanced model, by 0.81 pp on TCGA-UT and 1.39 pp on BRACS. After subtracting this balanced gain, it corrects 0.20 pp [−0.17, 0.57] of the imbalance on TCGA-UT and 1.20 pp [−0.16, 2.62] on BRACS. At native shares it is the only method that gains at least 1 pp (1.01 and 1.21 pp). On TCGA-UT, CUDA also lowers temperature-scaled negative log-likelihood (NLL) below that of CE, by 0.09 nats [0.07, 0.12] with Virchow2 and 0.15 nats [0.13, 0.17] with UNI2-h (99.84% intervals), and below that of the balanced CE model. The gain does not depend on adapting the encoder: with a frozen Virchow2, CUDA gains 1.83 pp over CE on BRACS and 0.89 pp on TCGA-UT, against 2.59 and 1.02 pp with LoRA. The best CUDA model on TCGA-UT is still the adapted one, by 2.77 pp (99.17% interval 2.13 to 3.38), because LoRA already raises CE by 2.64 pp there.

3. Mitigation does not fix calibration; one temperature does. The adapted CE models are overconfident even on balanced data, with raw ECE of 28–31 pp on BRACS and 17–18 pp on TCGA-UT across the two encoders. At $\rho = 100$, raw ECE with Virchow2 rises to 36.4 pp on BRACS and stays at 16.0 pp on TCGA-UT. Mitigation reduces this only in part. CUDA reduces it most, to 20.8 and 11.5 pp. cRT and balanced sampling can raise it. Without temperature scaling, every method stays at least 6.3 pp above the ECE of scaled CE. After scaling, as in Section 3, every method is as well calibrated as CE or slightly better, by at most 3.7 pp. Apart from CUDA, scaled NLL differs from that of scaled CE by −0.06 to +0.03 nats. CUDA's lower scaled NLL on TCGA-UT (finding 2) comes without lower ECE, so it reflects its accuracy gain rather than better calibration.

5.2 Limits of mitigation

1. Mitigation cannot add missing knowledge of a class. No method closes the gap on BRACS. The highest upper bound of the share intervals is 0.82 with Virchow2 and 0.71 with UNI2-h. The two tasks differ in what the imbalanced CE model still knows. On TCGA-UT it retains 61% tail recall at $\rho = 100$: the model has learned the rare classes, and the main problem is a misplaced decision boundary, which shifting the class scores can correct. On BRACS, tail recall falls to 5.1%, against 34.5% when balanced. Logit adjustment and GCL's head refit raise it only to 20–27%. This fits the sources of damage in Section 4.1. Rebalancing acts on the frequency effect but cannot supply missing tail support, which alone cost 3.29 pp on BRACS and 0.53 pp on TCGA-UT. **2. Mitigation has a cost.** Rebalancing shifts recall from head to tail: logit adjustment gives up about 4 pp of head recall on TCGA-UT and 9 pp on BRACS. Methods that change the head can also lower accuracy without imbalance. GCL and cRT lose 0.3 to 0.6 pp on the balanced allocation, and GCL loses up to 1.7 pp with UNI2-h on TCGA-UT. Their recovery over CE therefore understates how much imbalance they correct. Accounting for this balanced loss, GCL's head refit corrects about half of the damage on BRACS and about all of it on TCGA-UT. Logit adjustment and balanced sampling reduce to CE on balanced data and carry no such cost.

3. Mitigation matters only under severe imbalance. With the encoder adapted, native class shares cost little: 0.42 pp [0.06, 0.79] on TCGA-UT and 0.85 pp [−0.31, 2.08] on BRACS, against 1.18 and 4.05 pp for the frozen readout. On BRACS, this reflects a weaker balanced reference: the adapted balanced model is 3 pp less accurate than the frozen readout (36.53 against 39.57%), while

the imbalanced models are as accurate as the frozen readout. At native shares, only CUDA gains 1 pp; on TCGA-UT at $\rho = 50$, no method does.

Repeating the mitigation experiments with UNI2-h on the same patients and patches (on TCGA-UT at $\rho = 100$ only) leaves unchanged where each method acts. The same methods lift tail recall, post-hoc and train-time logit adjustment agree, and GCL's head refit corrects the most imbalance. Recovery point estimates change by less than 1 pp, but shares change with the damage. On TCGA-UT, UNI2-h's damage is 0.94 pp larger, so logit adjustment no longer recovers it fully (share 0.65 [0.54, 0.76]). On both tasks, CUDA's correction after subtracting its balanced gain becomes material with UNI2-h.

A Guide to the supporting experiments

The numbered folders below are under `experiments/03_class_imbalance/`. Each folder's `report/` directory contains a PDF and its LaTeX source. These reports provide the protocols, full tables, and uncertainty estimates for the results summarized here.

Headline results.

- **Imbalance curves and probability quality:** Reports 25 and 26, in `00_damage/25_damage_tcga_ut/report/` and `00_damage/26_damage_bracs/report/`, give the balanced baselines, damage at every ratio, native allocations, head-to-tail recall, and calibration before and after temperature scaling.
- **Class influence versus tail support:** Reports 27 and 28, in `01_cause/tcga_ut/27_prior_vs_support/report/` and `01_cause/bracs/28_prior_vs_support/report/`, show the paired frequency-only (called prior-only, P, in the reports) and support-only conditions, their interaction, and its uncertainty on each dataset.
- **Which classes become rare:** Report 29, in `01_cause/bracs/29_tail_assignment/report/`, rotates every BRACS class through the tail. Report 32, in `01_cause/tcga_ut/32_worst_assignment_rate/report/`, tests predicted damaging and mild class orders on TCGA-UT, including reversal of the most damaging order.
- **Encoder comparison:** Report 39, in `01_cause/cross/39_encoder_transfer/report/`, compares Virchow2 and UNI2-h on matched patients and patches and examines sensitivity to regularization selection.
- **Mitigation:** Report 40, in `02_mitigation/40_methods/report/`, defines the six methods and their taxonomy. Reports 41 and 43, in `02_mitigation/41_mitigation_bracs/report/` and `02_mitigation/43_mitigation_tcga_ut/report/`, measure each method's recovery on BRACS and TCGA-UT. Reports 42 and 44, in `02_mitigation/42_mitigation_uni2h/report/` and `02_mitigation/44_mitigation_tcga_ut_uni2h/report/`, repeat them with UNI2-h and test each difference between the encoders. Report 46, in `02_mitigation/46_ts_noninferiority/report/`, tests each method's temperature-scaled NLL against scaled CE with paired intervals.

Further reading. Reports 25 and 26 also track negative log-likelihood and fitted confidence corrections across the full imbalance grid. Reports 29 and 32 give class-level recall and confusion analyses behind the aggregate class-order effects, useful when the identity of a rare class matters. Report 39 adds per-class encoder comparisons and a fixed-regularization analysis, showing how the probe-selection rule affects the encoder contrast. Reports 41–43 add recovery at $\rho = 50$, and Reports 41–44 give recall by class rank, the selected method strengths, and probability quality per method.

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